



Regionally heterogeneous housing cycles and housing market stabilization policies: Evidence from Korea

Hyunduk Suh¹

Department of Economics, Inha University, Republic of Korea

ARTICLE INFO

JEL classification:

E32
E52
R31

Keywords:

Regionally heterogeneous housing cycles
Monetary policy
Macroprudential policy
Housing policy

ABSTRACT

Housing cycles can vary significantly across regions; this study investigates the macroeconomic implications of regionally heterogeneous housing cycles and stabilization policies. The general equilibrium model includes two separate regions and idiosyncratic shocks in regional housing markets. Simulations suggest that regional housing cycles can be a source of economic inequality between regions by asymmetrically affecting consumption, housing service, debt, and welfare. A regional housing boom caused by housing demand shocks improves the welfare of the borrowing agents in the region but reduces the other region's welfare. Conversely, a rise in housing prices caused by a low housing supply decreases the welfare of the agents in the region. Region-specific stabilization policies such as property tax, countercyclical loan-to-value (LTV), and housing supply policies can mitigate regional housing cycles. While the housing supply policy primarily benefits agents in the region that experiences the cycle, the LTV policy harms the welfare of borrowers in the region. The leaning against the wind monetary policy is relatively ineffective in stabilizing regional housing prices.

1. Introduction

Housing cycles are an important source of macroeconomic fluctuations. For example, [Leamer \(2007\)](#) suggests that residential investment is the best early warning signal of future recessions. [Iacoviello \(2005\)](#), [Iacoviello and Neri \(2010\)](#) suggest that the collateral channel associated with housing goods helps explain business cycle dynamics, and spillovers from housing markets exist and have become stronger over time.

Housing cycle variations may exist across regions in an economy large enough that it cannot be characterized as a single housing market. Panel (a) of [Fig. 1](#) displays the housing price indices of two regions in South Korea (hereafter, Korea): the Seoul Metropolitan Area and the Southeast Area.² The figure suggests the existence of regionally heterogeneous housing cycles in Korea, highlighted by two conspicuous periods that lasted for three to five years. From 2005 to 2008, the housing price index in the Seoul Metropolitan Area surged, while housing markets in the Southeast Area only mildly appreciated. Conversely, from 2010 to 2015, housing prices in the Southeast Area rose rapidly, while housing markets in the Seoul Metropolitan Area experienced a significant slump.

Various factors can cause these regionally heterogeneous housing cycles. Housing demand is affected by household income; thus, regional macroeconomic conditions causing household income to fluctuate differently across regions can be a source of regionally heterogeneous housing cycles. Another factor is changes in government policies that affect regional housing markets, such as large-scale land developments, which can influence regional housing supply. Additionally, shocks in housing-related taxes and mortgage lending regulations imposed with macroprudential purposes may vary across regions.

Panel (b) of [Fig. 1](#) indicates that regional housing cycles can be independent of regional macroeconomic cycles. From 2005 to 2015, although the macroeconomic cycles in the two regions were similar, housing cycles varied notably. This variation implies that other than regional macroeconomic conditions, idiosyncratic shocks originating from regional housing markets can be a source of regional housing cycles. Moreover, these idiosyncratic shocks can influence regional macroeconomic variables, such as residential investment and consumption, and eventually have welfare implications for economic agents.

Several economic questions may arise from these regionally heterogeneous housing cycles. For example, because of the positive propensity

E-mail address: hsuh@inha.ac.kr.

¹ I appreciate Hyunju Kang, Jaevin Park, the editor, the associate editor, and two anonymous reviewers of the *Economic Modelling* for their valuable suggestions. This work was supported by Inha University Research Grant, Republic of Korea.

² These regions are the most distinct large economic zones in Korea, accounting for 52 percent (Seoul Metropolitan) and 19 percent (Southeast) of domestic production, respectively, and 50 percent and 25 percent of the population, respectively, in 2019.

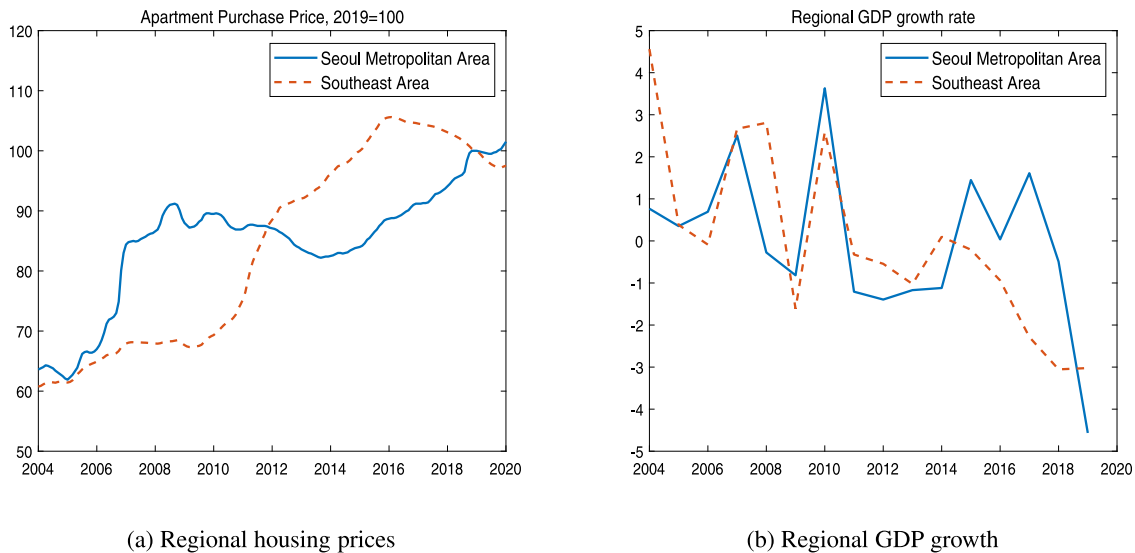


Fig. 1. Regional housing prices and GDP growth in Korea.

Note: In panel (a), the housing price indices are the Apartment Purchase Price, 2019=100. The index for the Southeast Area is the arithmetic average of the five first-tier administrative divisions in the region: Busan, Daegu, Ulsan, Gyeongsangnam province, and Gyeongsangbuk province. In panel (b), the regional gross domestic product (GDP) growth rate is mean-adjusted, as the number represents the deviation from the sample mean for each variable.

Source: KB Financial, Statistics Korea.

to consume out of housing wealth (Mian et al., 2013), a regional housing boom can create a regional consumption boom, adding regional heterogeneity into business cycles. Furthermore, an increase in household debt usually accompanies the housing boom. As household wealth and debt increase in the region, nontrivial implications may arise for welfare inequality between the agents that experience a housing boom and those in other regions. This problem can be even more complicated if a low housing supply in the region primarily or partially causes the increases in housing prices, indicating there are fewer housing goods to consume, affecting welfare negatively, and offsetting the positive contribution from the wealth effect. Moreover, governments can implement stabilization policies to alleviate overheating in regional housing markets, prevent the formation of housing or credit bubbles, and reduce regional inequality.

In this respect, this study primarily investigates the macroeconomic implications of the regionally heterogeneous cycles caused by idiosyncratic shocks in regional housing markets, such as Korea experienced. Specifically, we address the following questions. How do regional housing shocks affect regional output, consumption, and welfare? What are the effects of government policies such as tax, housing supply, macroprudential, and monetary policies in response to these shocks?

This study analyzes these questions by employing a dynamic stochastic general equilibrium (DSGE) model that features two regions across which consumption goods are tradable, but housing goods are attached to the regional markets. Agents in one region can invest in housing in the other region but cannot reside there. The model has other standard features in the New Keynesian literature: nominal rigidity, capital adjustment, and utilization costs. These features help the model fit data better and are useful for analyzing the effect of monetary policy on regional housing cycles. Model parameters are estimated using the Bayesian method and Korean data. Idiosyncratic shocks in regional housing supply or demand generate heterogeneous housing cycles. Specifically, we conduct a counterfactual experiment in which a series of idiosyncratic housing supply or demand shocks increases the housing price of one region by 15 percent. We analyze the response of the macroeconomic variables in both regions and how they are affected by the different use of policy instruments.

Simulating regional housing cycles suggests that they can affect regional housing production, debt, and consumption. When a housing

boom in region 1 is driven by a housing exuberance shock that represents the shock in housing demand, the region experiences a surge in housing production and an increase in consumption. The latter is mainly caused by the relaxation of borrowing constraints rather than the wealth effect. The shock can also cause inequality between regions by affecting welfare asymmetrically. The welfare of region 1 borrowers increases. Larger housing stocks provide more housing services, and borrowers receive additional welfare gains as they can use housing as collateral. In contrast, the welfare of all region 2 agents decreases. When region 1 housing prices increase via adverse housing supply shocks, this lowers the welfare of all region 1 agents. Decreased housing supply and housing services directly harm these agents. Conversely, the welfare of all region 2 agents increases.

The Korean government adopted region-specific housing market stabilization measures, such as property tax, countercyclical loan-to-value (LTV), and housing supply policies. The simulation suggests that those policies can mitigate regional housing cycles; however, it takes large policy responses if a housing demand shock causes the cycle. The LTV policy has a relatively weak impact on housing prices, as it only applies to new loans. Expansionary housing supply policy increases region 1 housing production, but demand policies, such as tax and LTV policies, slightly reduce it. Given supply or demand shocks, the welfare of region 1 agents consistently improves under the housing supply policy that provides more housing services to them. Additionally, the LTV policy consistently reduces the welfare of region 1 borrowers by tightening their credit constraints. “Leaning against the wind” monetary policy is relatively ineffective in stabilizing regional housing prices, as its effect is dispersed to housing markets in other regions and consumption. Higher interest rates during housing price appreciations negatively affect the welfare of borrowers in all regions.

1.1. Literature review

Previous studies confirm that housing cycles can vary across regions. Fratantoni and Schuh (2003) suggest that supply and demand in American regional housing markets may depend heavily on idiosyncratic and regional factors, providing evidence of regional heterogeneity in housing appreciation in the 1980s and 1990s. Del Negro and Otrok (2007) estimate a dynamic factor model on state-level housing price data and find that the local component mainly drove historical

housing price movements. A similar finding is reported by Fairchild et al. (2015), who estimate a dynamic factor model using the price/rent ratios of 23 major housing markets and find that local factors drive a large proportion of housing market volatility.

Several studies have highlighted the role of housing markets in macroeconomic fluctuations. Davis and Heathcote (2005) develop a DSGE model that incorporates the housing sector to match housing market dynamics with the business cycle patterns. Iacoviello (2005), Iacoviello and Neri (2010) use housing DSGE models to examine the propagation effect of housing markets on the macroeconomy. This study's model has some similar features to Iacoviello and Neri (2010), but we extend the model to include a multiregional setup with heterogeneous housing cycles and inter-regional housing investment. We also allow news shocks in housing markets to be sources of housing market fluctuations, as in Gomes and Mendicino (2015) and Ng and Feng (2016), who find a significant contribution of housing news shocks to the housing market and macroeconomic fluctuations. Moreover, we model household debt as a long-term amortizing mortgage loan, as in Alpanda and Zubairy (2017). Furthermore, this study has similarities to Lee and Song (2015) in estimating a housing DSGE model with Korean data and to Franz (2020) in discussing housing and borrower-related macroprudential policy tools in Korea.

This study aims to understand how regionally heterogeneous housing cycles impact the regional economy and welfare and how government policies affect those cycles. In this sense, the closest work in the literature is Funke et al. (2021), which examines the two-region DSGE model with regional housing markets. They conclude that the region-specific LTV policy performs better than the leaning against the wind monetary policy or national LTV policy in stabilizing the housing price variance. Although the model frameworks are similar, we focus on how regional housing cycles affect each agent in each region differently rather than the aggregate measures they study. Furthermore, this current study considers both housing supply and demand shocks. It compares different measures of region-specific stabilization policies, including the housing supply policy that has not been frequently investigated in the literature, while Funke et al. (2021) only analyzes the housing preference shock and region-specific LTV policy. Furthermore, this study is close to Basto et al. (2019) in studying the impact of LTV policy.

2. Housing market stabilization policies in Korea

When governments face regionally heterogeneous housing cycles, they can implement stabilization policies to alleviate overheating in regional housing markets, prevent the formation of speculative or credit-driven housing bubbles, and reduce regional inequality. This section summarizes policy measures for housing market stabilization used by the Korean government in recent years. Following Shin and Yi (2019), these measures are classified into tax, macroprudential, and housing supply policies. These policy instruments also have a commonality, as they can be applied region-specifically, notably in regions that experience a housing boom. Specifically, the Korean government designated policy target regions, distinguished by Speculation, Speculative Overheating, and Adjustment-required zones, depending on the degree of overheating. Each zone is subject to different degrees of tax and macroprudential regulations, while regulations on Speculation Zones are the strongest. Zone designations are lifted when the government determines that housing markets have stabilized. Fig. 2 shows the time-varying number of policy target zones in the Seoul Metropolitan Area. The figure shows the countercyclical nature of the policies, as the number of target zones increases during the 2016–2021 housing appreciation period but decreases in 2022 as housing prices start to fall. Also, Jung and Lee (2017) identify 15 changes in LTV limits in Korea during 2002–2014, approximately one change per year. Those changes are countercyclical; LTV limits are lowered during housing booms and vice versa during housing market downturns.

The following summarizes the details of each policy category.³ First, for housing market stabilization purposes, the Korean government introduced the Comprehensive Property Tax in 2005, levied on the owners of multi-housing and high-priced housing units in addition to regular property tax. The tax rate is progressive depending on the property value. In July 2020, the Korean government raised the Comprehensive Property Tax rate from 0.6–3.2 percent to 1.2–6.0 percent in response to sharp housing price appreciations. In addition, the capital gains tax rate for multi-housing owners increased by 20–30 percent.

Second, since 2002, the Korean government has implemented macroprudential regulations, such as LTV and debt-to-income (DTI) caps. Those macroprudential regulations are conducted countercyclically, leaning against the wind. Furthermore, different caps are applied across different policy target regions.⁴ For example, Table 1 summarizes region-specific LTV ratios as of May 2018. The lowest caps are applied in Speculation Zones, and regulations are less restrictive in Speculative Overheating and Adjustment-required zones. In 2019, the government further strengthened LTV regulations. For Speculative Overheating Zones, the LTV cap was lowered to 40 percent for housing units valued up to 900 million Korean won (KRW), then 20 percent for the proportion of the housing value exceeding 900 million KRW. Moreover, mortgage borrowing is even banned for houses over 1.5 billion KRW.

Third, the Korean government has implemented large-scale housing development plans and deregulated urban housing reconstruction procedures to boost the housing supply against rising housing prices. For example, in 2015, the government relaxed regulations on the minimum eligibility period for reconstruction, extending the grace period for three years to redeem excessive earnings from reconstruction, aiming to expedite the reconstruction of old housing to provide new housing. Furthermore, the government announced a long-term supply plan for 1.3 million units in the Seoul Metropolitan Area (15.2 percent of housing in the area as of 2019) in August 2020, followed by an additional 836,000 housing supply plan in February 2021. The plan includes the development of new cities around Seoul, mainly by lifting regulations on nearby urban greenbelt areas that were kept undeveloped for environmental preservation purposes.

3. Model

The key elements of the model are close to the one-country housing DSGE model proposed by Iacoviello and Neri (2010) and the two-region housing DSGE model by Funke et al. (2021). The two regions are indexed by $j = 1, 2$. Consumption goods are assumed to be tradable between the regions; thus, a single price applies (i.e., goods inflation is the same in both regions). In contrast, housing goods and labor are attached to the regional markets⁵; thus, housing prices and wages might differ between regions.

Inter-regional housing investment is allowed; agents in one (“home”) region can purchase housing units in the other (“external”) region. In reality, inter-regional housing investment accounts for a significant proportion of housing purchases. The Korean Homeownership Statistics in 2019 suggests that agents outside the region own approximately 14 percent of housing. Although these inter-regional housing investors cannot reside in their external housing units, they receive rental fees from them. Moreover, these inter-regional housing assets can be pledged as collateral for borrowing.

Both regions have identical population and economic structures, except for regionally idiosyncratic housing shocks, implying that both regions have a symmetric steady state in which the inter-regional trade balance is zero.

³ Financial Stability Reports, a regularly published report by the Bank of Korea, offers more details of current operations of housing-related macroprudential policies in Korea.

⁴ For details of those policy changes, please see Jung and Lee (2017).

⁵ The model abstracts from immigration between regions, focusing on the short-to-medium term effect caused by a housing cycle.

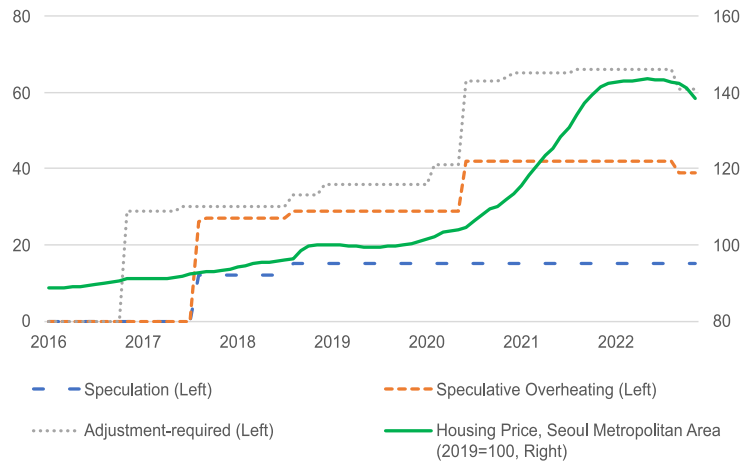


Fig. 2. Number of housing market stabilization policy target zones, Seoul Metropolitan Area.

Note: “Speculation”, “Speculative Overheating”, and “Adjustment-required” means the number of each target zone in the Seoul Metropolitan Area, respectively. The housing price index is the apartment purchase price for the Seoul Metropolitan Area, 2019=100.

Source: Korean Ministry of Land, Infrastructure, and Transport, KB Financial.

Table 1

Region-specific macroprudential regulations in Korea.

Source: Shin and Yi (2019).

Zones	Major regulations	Areas
Speculation	LTV/DTI requirements (40/40%) Limit on the number of mortgage loans	11 districts in Seoul, Sejong
Speculative Overheating	LTV/DTI requirements (40/40%) Tighter screening of the financing methods for new housing purchases	All 25 districts in Seoul, Sejong, 1 city in Gyeonggi, 1 district in Daegu
Adjustment-required	LTV/DTI requirements (60/50%)	All 25 districts in Seoul, Sejong, 7 cities in Gyeonggi, 7 districts in Busan

Note: This table summarizes region-specific housing-related macroprudential regulations in Korea as of May 2018.

3.1. Households

Households receive utility from consumption and housing goods, and disutility exists in labor, which they supply to both the consumption and the housing goods production sectors. Each region has patient households (“savers”) and impatient households (“borrowers”) distinguished by the discount factor. Patient households have a higher discount rate (β) than impatient households (β'), and they are net savers who supply credit in the economy in the steady state and its neighborhood. They also own capital, make capital investment decisions, and make four choices about housing: purchasing housing units in the home and external regions and residing in rental units in the home region owned by the external patient and impatient households. Housing units in the home region and rental units provide housing services.⁶ As mentioned, the external housing units they purchase as inter-regional housing investments do not provide housing services but do provide rent. Abstracting from the regional subscript j , patient households in each region maximize

$$E_0 \sum_{t=0}^{\infty} (\beta G_c)^t z_t \left[\ln(c_t - \epsilon c_{t-1}) + J \ln h_t + J^R \{ \ln h_t^R + \ln h_t'^R \} - \frac{\tau_t}{1+\eta} (n_{c,t}^{1+\psi} + n_{h,t}^{1+\psi})^{\frac{1+\eta}{1+\psi}} \right], \quad (1)$$

where c_t is consumption, h_t is the self-owned housing units in the home region, and h_t^R and $h_t'^R$ are the rental units in the home region

⁶ Both owner-occupied and rental housing services affect utility. One can think of a large family with members residing separately in different housing types.

owned by the external patient and impatient households, respectively. $n_{c,t}$, $n_{h,t}$ denote the labor supply for the consumption and housing goods production sectors, respectively. G_c is the trend growth, z_t is the time preference shock, ϵ is a parameter governing the degree of habit formation, and τ_t is the labor preference shock. J represents the relative degree of preference for housing services from self-owned units, and J^R represents that from rental units.

If we denote the variables related to external housing as superscript A , patient households period budget constraint is given as

$$c_t + d_t + \frac{k_{c,t}}{A_{k,t}} + k_{h,t} + q_t h_t + q_t^A h_t^A + r_t^h h_t^R + r_t^{h'} h_t'^R = \frac{R_{t-1} d_{t-1}}{\pi_t} + \left(R_{c,t} z_{c,t} + \frac{1 - \delta_{kc}}{A_{k,t}} \right) k_{c,t-1} + (R_{h,t} z_{h,t} + 1 - \delta_{kh}) k_{h,t-1} + q_t (1 - \delta_h - \tau_t^h) h_{t-1} + q_t^A (1 - \delta_h - \tau_t^{h,A}) h_{t-1}^A + w_{c,t} n_{c,t} + w_{h,t} n_{h,t} + r_t^{h,A} h_t^A + Div_t + T_t - \phi_t - \frac{a(z_{c,t}) k_{c,t-1}}{A_{k,t}} - a(z_{h,t}) k_{h,t-1}. \quad (2)$$

where d_t is saving, $k_{c,t}$, $k_{h,t}$ are the capital for the consumption and housing goods production sectors, respectively. h_t^A denotes the ownership of housing goods located in the external region. q_t and q_t^A are the housing prices in the home and external regions, respectively, r_t^h and $r_t^{h'}$ are the housing rents paid to the external patient and impatient households, and $r_t^{h,A}$ is the rent received from external tenants. $R_{c,t}$, $R_{h,t}$, $z_{c,t}$, $z_{h,t}$, $w_{c,t}$, and $w_{h,t}$ denote the returns on capital, utilization rates, and real wages of the consumption and housing production sectors.

Div_t includes lump-sum dividends from firms and labor unions, and T_t is the government transfer minus the lump-sum tax. The transfer comes from property tax revenue, and the lump-sum tax is used

to finance government spending. Once the amount of government spending (G_t) is determined, α fraction of the spending is assumed to be financed from saving households, where α is the share of saving households in the production function. The remaining fraction is taxed from borrowing households.

τ_t^h and $\tau_t^{h,A}$ denote the time-varying property tax rates for home and external housing, respectively. ϕ_t is the capital adjustment cost and $a(z_{c,t})$ and $a(z_{h,t})$ are the capital utilization costs. $A_{k,t}$ is the investment-specific shock, which only applies to the capital used for consumer goods production.

The patient households' optimal choices concerning owner housing units in the domestic and external regions are, respectively,

$$q_t \lambda_t = \frac{J z_t}{h_t} + \beta G_c u_{q,t} E_t q_{t+1} \lambda_{t+1} (1 - \delta_h - \tau_t^h), \quad (3)$$

and

$$q_t^A \lambda_t = r_t^{h,A} \lambda_t + \beta G_c u_{q,t}^A E_t q_{t+1}^A \lambda_{t+1} (1 - \delta_h - \tau_t^{h,A}). \quad (4)$$

λ_t stands for the Lagrangian multiplier on the budget constraint that also equals the marginal utility of consumption. $u_{q,t}$ and $u_{q,t}^A$ are housing exuberance shocks to the future expectation of housing prices in the home and external regions, respectively. This housing exuberance shock is introduced by [Alpanda and Zubairy \(2017\)](#); it captures the pure expectational elements in agents' exuberance regarding future house price movements, which creates a wedge between the observed house price and underlying fundamental value. Housing demand fluctuation can alternatively be captured by the housing preference shock, that is, time-varying changes in housing preference. The difference is that the housing exuberance shock represents a deviation from rational expectations, while the housing preference shock captures the change in preferences in line with rational expectations. In the presence of the former shock, macroprudential policies have the additional role of correcting irrational exuberance while reducing externality during the boom–bust cycles. Furthermore, the former shock only indirectly affects welfare through the changes in housing prices, while the latter directly affects welfare. Appendix A.5 summarizes posterior estimates of deep parameters and model moments estimated using the housing preference shock. In terms of model performance, those two shocks yield similar results.

The left-hand side of Eq. (3) implies the foregone utility from purchasing an additional housing unit, while the right-hand side represents marginal utility from that housing unit plus discounted future utility from the additional saving in housing. The implication of Eq. (4) is similar, except that households receive rent $r_t^{h,A}$ instead of utility from external housing purchased.

The optimality conditions for the rental housing units are given as

$$J^R \frac{z_t}{h_t^R} = r_t^h \lambda_t, \quad (5)$$

for the rental units owned by external patient households, and

$$J^R \frac{z_t}{h_t^R} = r_t^{h'} \lambda_t, \quad (6)$$

for the rental units owned by external impatient households. Equations (5) and (6) imply that the marginal utility from rental housing services should equal the utility value of the rent that patient households pay to external homeowners. The other optimality conditions of the patient households are similar to [Iacoviello and Neri \(2010\)](#) and are summarized in the Appendix.

Impatient households have a lower discount factor than patient households ($\beta' < \beta$); thus, their financial position is that of a net borrower in the steady state and its neighborhood. Because of information asymmetry, collateral is required for borrowing, and housing goods can serve as collateral. Because impatient households need housing goods as collateral, we assume they are homeowners by nature and do not reside in rental units.⁷ In addition, they can make inter-regional housing

investments by purchasing external housing units. These transregional housing assets can be pledged as collateral for further borrowing. They maximize

$$E_0 \sum_{t=0}^{\infty} (\beta' G_c)^t z_t \left[\ln(c'_t - \epsilon c'_{t-1}) + J \ln h'_t - \frac{\tau_t}{1 + \eta'} (n'^{1+\psi'}_{c,t} + n'^{1+\psi'}_{h,t})^{\frac{1+\eta'}{1+\psi'}} \right], \quad (7)$$

subject to the budget constraint

$$c'_t + q_t h'_t + q_t^A h'^A_{t-1} + \frac{R_{t-1} d'_{t-1}}{\pi_t} = \frac{w_{c,t} n'_{c,t}}{X_{wc,t}} + \frac{w_{h,t} n'_{h,t}}{X_{wh,t}} + d'_t + q_t (1 - \delta_h - \tau_t^h) h'_{t-1} + q_t^A (1 - \delta_h - \tau_t^{h,A}) h'^A_{t-1} + r_t^{h'} h'^A_{t-1} + Div'_t + T'_t. \quad (8)$$

The left-hand side of the above budget constraint consists of consumption, domestic housing, external housing, and debt carried from the previous period ($\frac{R_{t-1} d'_{t-1}}{\pi_t}$). The right-hand side consists of labor income from the consumption and housing goods production sectors, new debt (d'_t), the return value of the previous period's domestic and external housing, rent received from external tenants, dividends, and transfers minus the lump-sum tax.

The collateralized debt and borrowing constraint design follows the amortizing structure of [Alpanda and Zubairy \(2017\)](#). A certain proportion of the debt balance is repaid in every period; thus, the law of motion for real debt is given as $d'_t = loan'_t + (1 - \kappa_m) d'_{t-1} / \pi_t$, where d'_t is the level of debt, $loan'_t$ is new loans, and κ_m denotes the amortization rate of the debt per period. There is a [Kiyotaki and Moore \(1997\)](#)-type borrowing constraint for acquiring new housing. Specifically, the size of new loans is constrained by the LTV limits (m_t for home housing and m_t^A for external housing) times new housing acquisition, plus a certain proportion of home equity extraction (γ_m), where home equity is defined as the value of housing minus the value of the remaining debt:

$$loan'_t \leq m_t [q_t (h'_t - (1 - \delta_h) h'_{t-1})] + m_t^A [q_t^A (h'^A_{t-1} - (1 - \delta_h) h'^A_{t-1})] + \gamma_m \left[q_t (1 - \delta_h) h'_{t-1} + q_t^A (1 - \delta_h) h'^A_{t-1} - (1 - \kappa_m) \frac{d'_{t-1}}{\pi_t} \right]. \quad (9)$$

The borrowers' optimal choice of debt gives rise to the Euler equation that accounts for the borrowing constraint:

$$\lambda'_t - \mu_t = \beta' G_c E_t \left[\lambda'_{t+1} \frac{R_t}{\pi_{t+1}} - \frac{\mu_{t+1}}{\pi_{t+1}} (1 - \kappa_m) (1 - \gamma_m) \right], \quad (10)$$

where λ'_t is the Lagrangian multiplier associated with the budget constraint and μ_t is the Lagrangian multiplier associated with the borrowing constraint (9), where $loan'_t$ is substituted out by using the law of motion for debt.

Because impatient households receive utility from domestic housing and rent from external housing, the optimality condition for housing goods can be derived as

$$q_t (\lambda'_t - \mu_t m_t) = \frac{J z_t}{h'_t} + \beta' G_c u_{q,t} E_t q_{t+1} [\lambda'_{t+1} (1 - \delta_h - \tau_t^h) - \mu_{t+1} (m_{t+1} - \gamma_m) (1 - \delta_h)], \quad (11)$$

for domestic housing, and

$$q_t^A (\lambda'_t - \mu_t m_t^A) = r_t^{h'} \lambda'_t + \beta' G_c u_{q,t}^A E_t q_{t+1}^A [\lambda'_{t+1} (1 - \delta_h - \tau_t^{h,A}) - \mu_{t+1} (m_{t+1}^A - \gamma_m) (1 - \delta_h)], \quad (12)$$

for the external housing.

⁷ It is possible to relax this assumption and allow impatient households to live in rental units; this does not affect the paper's key results.

3.2. Production

In each region, j , a representative intermediate goods producer for consumption goods uses capital and the combined labor of patient and impatient households as production inputs. Its production technology is given as

$$y_{j,t} = (A_{c,t} n_{j,c,t}^\alpha n_{j,c,t}^{1-\alpha})^{1-\mu_c} (z_{c,t} k_{j,c,t-1})^{\mu_c}, \quad (13)$$

where $k_{j,c,t-1}$ denotes the capital used for consumption goods production in the j th region.

Production technology for housing goods is given as

$$IH_{j,t} = (A_{j,h,t} A_{j,h,t-8}^\alpha n_{j,h,t}^\alpha n_{j,h,t}^{1-\alpha})^{1-\mu_h-\mu_l} (z_{h,t} k_{j,h,t-1})^{\mu_h} l^{\mu_l}. \quad (14)$$

In Eq. (14), $A_{j,h,t}$ is a technology shock in housing production that can vary between regions. $A_{j,h,t-8}^\alpha$, referred to as the housing supply policy hereafter, captures the government regulations and other policy instruments on housing production (supply) that also apply region-specifically. It includes regulations on land to be developed as housing complexes, regulations on urban housing construction and reconstruction, provision of urban infrastructure that improves the profitability of housing construction, and government incentives for housing developers. Because it usually takes some time for these measures to be implemented, we assume a time lag of eight periods for those policy measures to be effective.⁸

$k_{j,h,t-1}$ denotes the capital used for housing goods production in the j th region, and l is land; the size of l is normalized to 1. The profit maximization of intermediate goods producers and housing goods producers for different types of labor and capital determines the input demand for each type of good in each region. The Appendix summarizes the first-order conditions from the optimal decisions made by production firms.

3.3. Nominal rigidity

Nominal rigidity exists in the price determination of consumption goods and wages. The form of price and wage rigidity in our model largely follows (Gerali et al., 2010), with slight modifications. Intermediate goods are purchased by retailers who produce differentiated products in a monopolistic competitive environment. Those differentiated goods are packaged for use in the final consumption of households. Retailers can optimally choose the price of their product given its demand schedule, but they face a quadratic cost in adjusting the price à la (Rotemberg, 1982). In Appendix A.2, we provide formal descriptions of this nominal rigidity.

3.4. Market clearing

The market-clearing condition for consumption goods is that total production in the economy equals total expenditure, including consumption, capital investment, government spending (g_t), and adjustment costs ($adj_{j,t}$) for capital, price, and wage. Government spending is exogenous and evenly distributed across regions:

$$\sum_{j=1}^2 y_{j,t} = \sum_{j=1}^2 \left(c_{j,t} + c'_{j,t} + \frac{1}{A_{k,t}} (k_{j,c,t} - (1 - \delta_{kc}) k_{j,c,t-1}) + k_{j,h,t} - (1 - \delta_{kh}) k_{j,h,t-1} + \frac{g_t}{2} + adj_{j,t} \right). \quad (15)$$

Housing markets clear for each region; the sum of the owner and rental housing units equals new housing installed plus the remaining value of those units from the previous period:

$$h_{j,t} + h'_{j,t} + h_{j,t}^R + h_{j,t}^{R'} = IH_{j,t} + (1 - \delta_h) [h_{j,t-1} + h'_{j,t-1} + h_{j,t-1}^R + h_{j,t-1}^{R'}] \quad (16)$$

for $j = 1, 2$.

Following Iacoviello and Neri (2010), GDP in this economy is defined as the sum of consumer goods production and new housing installed:

$$GDP_t = \sum_{j=1}^2 (y_{j,t} + q_{ss} IH_{j,t}). \quad (17)$$

3.5. Monetary policy

The central bank adjusts the nominal interest rate according to the Taylor rule. The monetary policy responds systemically to consumer price index inflation and GDP growth:

$$R_t = R_{t-1}^{r_R} \pi_t^{(1-r_R)r_\pi} \left(\frac{GDP_t}{G_c GDP_{t-1}} \right)^{(1-r_R)r_Y} \bar{r}^{1-r_R} u_{R,t}, \quad (18)$$

where R_t is the gross interest rate and $u_{R,t}$ is an exogenous shock in the interest rate.

3.6. Housing market stabilization policy

Governments can implement stabilization policies to alleviate overheating in regional housing markets when facing regionally heterogeneous housing cycles. As mentioned earlier, the Korean government implemented policy instruments classified as tax, macroprudential, and housing supply policies. Considering the countercyclical nature of those tools, we assume that the tax rate $\tau_{j,t}^h$, LTV ratio $m_{j,t}$, and the housing supply policy $A_{j,h,t}^\alpha$ in each region are characterized by the following equations, as they systemically respond to regional housing prices:

$$\tau_{j,t}^h = \tau_{ss}^h \left(\frac{q_{j,t}}{q_{ss}} \right)^{\kappa_\tau}, \quad j = 1, 2. \quad (19)$$

$$m_{j,t} = m_{ss} \left(\frac{q_{j,t}}{q_{ss}} \right)^{-\kappa_{ltv}}, \quad j = 1, 2. \quad (20)$$

$$A_{j,h,t}^\alpha = \left(\frac{q_{j,t-2}}{q_{ss}} \right)^{\kappa_{hs}}, \quad j = 1, 2. \quad (21)$$

In Eq. (21), we allow a time lag of two periods considering housing supply plans take some time to develop.

3.7. Inter-regional debt and risk premium

Because a nonzero inter-regional trade balance per period may exist, inter-regional debt also exists. Inter-regional debt increases when total expenditure in the region exceeds total production. The steady-state value of debt is zero, and the law of motion for debt d_t^{IR} is given as

$$d_t^{IR} = \frac{R_{t-1} d_{t-1}^{IR}}{\pi_t} + y_{1,t} + q_{1,t} IH_{1,t} - \left(c_{1,t} + c'_{1,t} + \frac{1}{A_{k,t}} (k_{1,c,t} - (1 - \delta_{kc}) k_{1,c,t-1}) + k_{1,h,t} - (1 - \delta_{kh}) k_{1,h,t-1} + \frac{g_t}{2} + adj_{1,t} \right) - q_{1,t} [h_{1,t} + h'_{1,t} - (1 - \delta_h) (h_{1,t-1} + h'_{1,t-1})] - q_{2,t} [h_{2,t}^R + h_{2,t}^{R'} - (1 - \delta_h) (h_{2,t-1}^R + h_{2,t-1}^{R'})]. \quad (22)$$

According to the above definition, a positive (negative) value of d_t^{IR} implies that region 2 (1) is a net debtor.

We assume that the level of inter-regional debt affects the region's risk premium and interest rate. This condition is necessary to rule out an equilibrium in which one region accumulates persistent debt. This situation can also be seen as an analog of the exchange rate

⁸ Key results of the paper are robust when shorter time lags are considered.

risk premium, which depends on the net foreign debt in small open economy models, as in [Schmitt-Grohé and Uribe \(2003\)](#) and [Christiano et al. \(2011\)](#). Specifically, we assume that the interest rate in each region is determined as follows:

$$\hat{R}_{1,t} = \hat{R}_t, \quad \hat{R}_{2,t} = \hat{R}_t + \phi^R d_t^{1R}, \quad (23)$$

where $\hat{R}_{j,t}$ is the interest rate in region j in logarithm. We choose the value of ϕ^R , which represents the sensitivity of the risk premium to debt in region 2, to be very small. This value is based on the assumption that nationally operating banks only marginally discriminate between the two regions based on inter-regional debt.

3.8. Exogenous process

There are two regionally idiosyncratic housing market shocks: the housing exuberance shock $u_{j,q,t}$ and the shock in housing productivity $A_{j,h,t}$, which is denoted as $\epsilon_{j,h,t}$. Hereafter, we use “housing demand shock” and “housing supply shock” interchangeably, as the former represents the shock in housing demand and the latter represents the shock in housing supply. Furthermore, we add a news shock to the housing supply and assume that the process of housing productivity is $A_{j,h,t} = \rho_h A_{j,h,t-1} + \epsilon_{j,h,t} + \epsilon_{j,h,t-4}^n \cdot \epsilon_{j,h,t-4}^n$ is the news shock that conveys information on the housing supply realized after four quarters.

Other exogenous shocks, general preference z_t , labor preference τ_t , total factor productivity $A_{c,t}$, capital efficiency $A_{k,t}$, monetary policy $u_{R,t}$, price markup ϵ_t^p , and government spending g_t , are commonly applied to both regions. All shocks, except the monetary policy shock, follow the AR(1) process in log. The monetary policy shock is assumed to be independent and identically distributed.

4. Parameter calibration and estimation

We calibrate and estimate the model parameters using macroeconomic time-series data in Korea. This section discusses the details of the calibration and estimation procedures.

4.1. Calibration

Some of the model parameters are calibrated. We use the calibration values of [Iacoviello and Neri \(2010\)](#) for the savers' discount factor, depreciation rates, and the share of capital, land, and intermediate goods in housing production. Considering the lower risk premium during the sample period, we set the borrowers' discount factor at 0.98, higher than the 0.97 of [Iacoviello and Neri \(2010\)](#). ϵ , the habit formation parameter, and α , the share of saving households in the production function, are calibrated at 0.7 and 0.8, respectively. These values are consistent with the posterior mean values of [Iacoviello and Neri \(2010\)](#). The values of μ_c (the capital share in consumption goods production) and J (the relative importance of housing services in household preferences) are chosen to match the sample consumption/GDP ratio (0.56) and residential investment/GDP ratio (0.04) in Korea. We also calibrate the government spending/GDP ratio to match the sample value of 0.158.

The steady-state LTV ratio is set to 0.8. The home equity withdrawal rate γ_m and debt amortization rate κ_m are chosen to be close to the calibrated values presented by [Alpanda and Zubairy \(2017\)](#). J^R , the relative preference for rental housing, is set to 0.0083. This value implies that outside agents own approximately 14 percent of housing in each region, matching the value of the Korean Homeownership Statistics in 2019. We choose a small value of 0.01 for ϕ^R , the sensitivity of the regional risk premium to inter-regional debt, so that inter-regional debt is negligible in the regional interest rate. τ_{ss}^h is set to 0.0025, implying that the steady-state property tax rate is approximately 1 percent when annualized. Table A.1 in the Appendix summarizes the definitions and values of the calibrated parameters.

4.2. Estimation

The remaining model parameters are estimated using the Bayesian method. Because regional income accounting data are not available at a quarterly frequency, the estimation is conducted using national data. This strategy also fits our analysis because we assume both regions have identical economic structures, except for regionally idiosyncratic housing shocks. This approach means the structural parameters in both regions are the same.

We use nine observable variables from Korean data: consumption, residential investment, nonresidential investment, government spending, the GDP deflator, labor hours in the goods sector, the three-month certificate of deposit interest rate, the wage, and the housing price index. Among these, all the variables except the interest rate are log-differenced. As in [Iacoviello and Neri \(2010\)](#), consumption, investments, and government spending are normalized to per capita terms. The GDP deflator translates the nominal wage and housing price index into real values. We allow nonresidential investment and housing prices to have different trends from GDP growth.

The data source for national income statistics and interest rates is the Bank of Korea Economic Statistics System, while wage and labor hours are from the Ministry of Employment and Labor. The housing price index is obtained from the Korea Real Estate Board, and the sample period is 2000:1Q–2019:3Q.

Tables A.2 and A.3 in the Appendix report the prior distribution, posterior mean, and 90 percent highest posterior density interval obtained from the Markov Chain Monte Carlo draws. The prior distributions for the structural parameters and parameters governing the shock processes are mostly chosen to be consistent with those assumed by [Iacoviello and Neri \(2010\)](#). The parameters for capital, price, and wage adjustment costs have a gamma prior with a mean of 10 and a standard deviation of 0.25. The priors for the inverse Frisch elasticity of labor η and inverse elasticity of sectoral labor substitution ψ are a gamma distribution and normal distribution, respectively. Following the standard approach in the literature, the priors for the AR(1) coefficients and shock volatility of the exogenous processes have a beta distribution and an inverse gamma distribution, respectively.

We linearize the model equations around the steady state to obtain the posterior distribution and simulate 300,000 sets of the posterior parameters using the Metropolis–Hastings algorithm. Excluding the burn-in period, the last 180,000 draws are used to formulate the posterior distributions. [Geweke \(1992\)](#)'s convergence statistics support the convergence of the posterior distribution of all parameters.

Table A.4 (a) in the Appendix compares the key moments implied by the estimated model with the moments of sample data. Generally, the model exhibits higher volatility than the sample, but the difference is insignificant. Furthermore, the model closely follows the relative volatility pattern in the sample data, for example, high volatility in residential and nonresidential investments and low volatility in consumption. In addition, the model generates similar correlation patterns between the key variables to those in the sample data. For example, consumption positively correlates with nonresidential investment, wages, and housing prices but negatively correlates with inflation. The only notable difference between the model and the sample is the correlation between housing prices and residential investment. The model reports a high correlation (0.54) between these two variables, suggesting that demand shocks primarily drive the housing market; however, the sample has a correlation value close to zero. While [Fehrlé \(2019\)](#) suggests that housing capital utilization and adjustment cost in the model can generate a positive correlation between housing prices and residential investment, as reported here, the sample correlation value near zero is indeed peculiar in Korea. It mainly arises from the negative correlation between two variables from 2010 to 2019: –0.31. [Oh \(2018\)](#) explains this negative correlation because residential investment during this period is driven by supply factors rather than demand factors. The housing supply led by large-scale public land

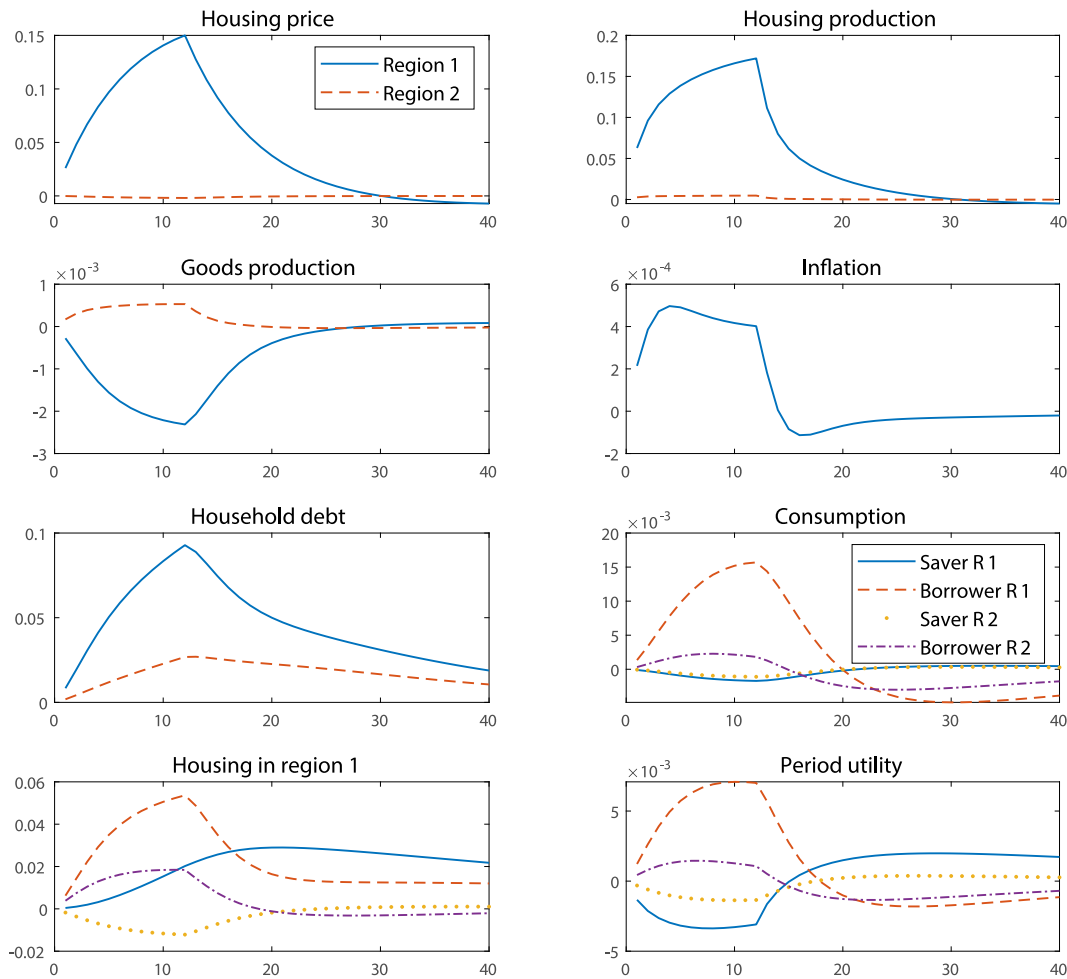


Fig. 3. Regional housing cycle in region 1, housing exuberance shock.

Note: The figure presents the response of each variable to 12 consecutive unanticipated housing exuberance (demand) shocks that raise housing prices in region 1 by 15 percent. All the variables, except period utility, are expressed as log deviations from the steady state; period utility is expressed as the percentage change from the steady state.

development projects (which coincides with the housing supply policy mentioned later) is the primary driver of residential investment, providing downside pressure on housing prices while residential investment increases.

Table A.4 (b) reports the key variables' unconditional forecast error variance decomposition. Housing exuberance shocks explain about 30 percent of the variation in real housing prices; various shocks, such as preference, labor supply, productivity in the consumption goods sector, and price markup shocks, explain the rest. Conversely, housing supply and news shocks contribute only a small proportion, jointly accounting for only 2.7 percent of the error variance in the housing price forecast. Housing exuberance shocks also explain approximately 25 percent of the variation in residential investment. Housing supply shocks (43 percent) and housing news shocks (4 percent) make a more considerable contribution to residential investment than their respective contributions to housing prices. Regarding other shocks, labor supply shocks explain approximately 40 percent of the consumption variation and 23 percent of the inflation variation, and demand preference shocks account for approximately 43 percent of the inflation variation and 20 percent of the consumption variation.

5. Results

5.1. Macroeconomic effects of regionally heterogeneous housing cycles

We conduct a counterfactual experiment to analyze the macroeconomic implications of regionally heterogeneous housing cycles. A

series of housing demand or supply shocks generate a regional housing cycle. To create a housing boom driven by housing demand shocks, we assume that only region 1 is hit by positive, unanticipated housing exuberance shocks of identical size for 12 consecutive quarters. The shock's size is chosen so that these shocks increase housing prices in region 1 by 15 percent at their peak. This simulation is conducted by the first order approximation of the model around the steady state. Thus, the result reported here is accurate up to that degree. The quantitative results may become inaccurate if shocks cause the economy to deviate far from the steady state.

Fig. 3 summarizes the responses of the key macroeconomic variables for the 10 years after the beginning of the cycle. Housing prices in region 1 rise as long as shocks hit the economy (i.e., for three years) and then decline. Increasing housing demand in region 1 also causes regional housing production to increase by approximately 17 percent of its steady-state value. Housing prices in region 2 are almost unchanged. The production of consumption goods falls as capital in region 1 is directed to the more profitable housing production sector. In region 1, the production of consumption goods decreases by approximately 0.2 percent at the cycle's peak. This decrease is partially offset by the increase in production in region 2. Inflation rises, but only by less than 0.1 percent.

During the housing boom, household debt in region 1 continually increases to about 9 percent of the steady-state value. Household debt in region 2 also increases but to a lesser degree. Higher credit allows borrowing households in both regions to enjoy larger consumption.

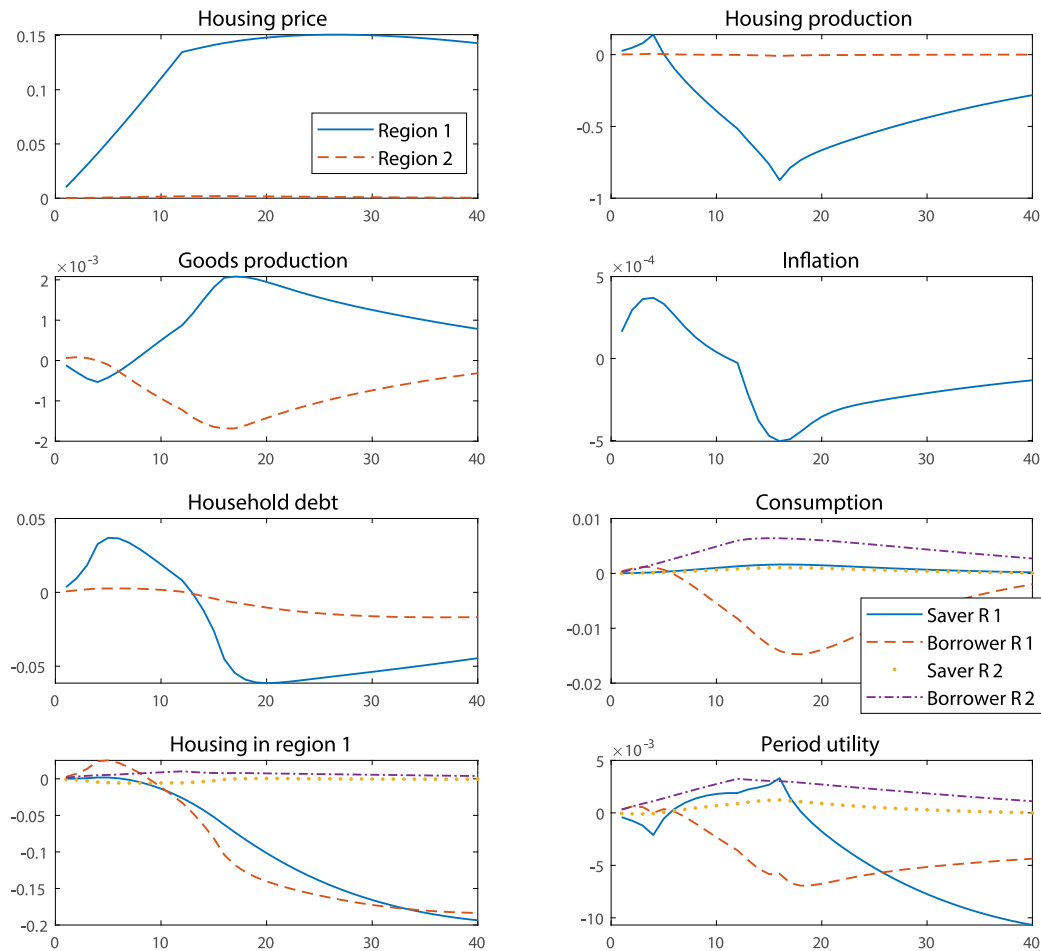


Fig. 4. Regional housing cycle in region 1, housing supply news shock.

Note: The figure presents the response of each variable to the 12 consecutive unanticipated housing productivity (supply) news shocks that raise housing prices in region 1 by 15 percent. All the variables, except period utility, are expressed as log deviations from the steady state; period utility is expressed as the percentage change from the steady state.

Borrowers in region 1 increase consumption by approximately 1.6 percent, greater than 1 standard deviation of quarterly consumption in data. Those in region 2 increase consumption by a minor degree. Conversely, the consumption of saving households decreases in both regions. Therefore, region 1 experiences a consumption boom driven more by the relaxation of credit constraints than the wealth effect.

When housing prices rise, borrowers in region 1 are also the largest purchasers of region 1 housing; however, their housing stock falls after the housing boom's peak. Conversely, savers in region 1 steadily accumulate region 1 housing even after the housing cycle passes its peak. Inter-regional housing purchases from region 2, representing mostly speculative motives, increase from borrowers who have an additional motive to expand their credit constraints but decrease from savers.

Fig. 3 also shows the change rate of period utility from its steady-state value, suggesting that region 1 borrowers enjoy the largest utility gain during the regional housing boom caused by the housing demand shock, especially during early periods. The housing boom increases their borrowing constraint to increase their consumption and housing services. Region 1 savers' utility falls during early periods but increases later as their housing stock accumulates. Utility changes of savers and borrowers in region 2 are smaller than those of region 1 agents. Overall, borrowers in both regions are better off during early periods when housing prices rise; however, their utility falls during later periods when housing prices decline. The results suggest that regional housing cycles can be a source of economic inequality between regions and the level of financial status. Furthermore, borrowers in other regions, who are also inter-regional housing investors, are crucial in the spillover of the regional housing cycle to other regions.

Fig. 4 summarizes the response of the macroeconomic variables during a regional housing price appreciation caused by adverse housing productivity news shocks. Again, the shocks hit only region 1 for 12 consecutive quarters. The qualitative and quantitative results are similar when we use the contemporaneous shock in housing productivity instead of the news shock; thus, we omit the result from the contemporaneous housing productivity shock for brevity. The shock's size is chosen so that the shocks raise housing prices in region 1 by 15 percent at their peak.

The rise in housing prices is slower than the rise induced by the housing exuberance shock; however, housing prices remain high for extended periods because it requires a long time for the reduced housing supply to recover. Housing production in region 1 initially increases until negative news materializes, but it declines sharply because of reduced productivity. Housing prices and production in region 2 are barely affected. Goods production from region 1 initially decreases; however, it increases as housing production declines because the lower return on capital in the housing production sector induces more investment in the consumption goods sector. This increase is offset by the decrease in goods production in region 2. The response to inflation is less than 0.05 percent.

In contrast to the housing demand shock, household debt in region 1 initially increases; however, it later decreases because of the reduced housing stock to be pledged as collateral. Higher initial credit allows region 1 borrowers to increase consumption and housing stocks in the early periods, but falling housing supply and tightened credit constraints cause them to reduce consumption and housing in the

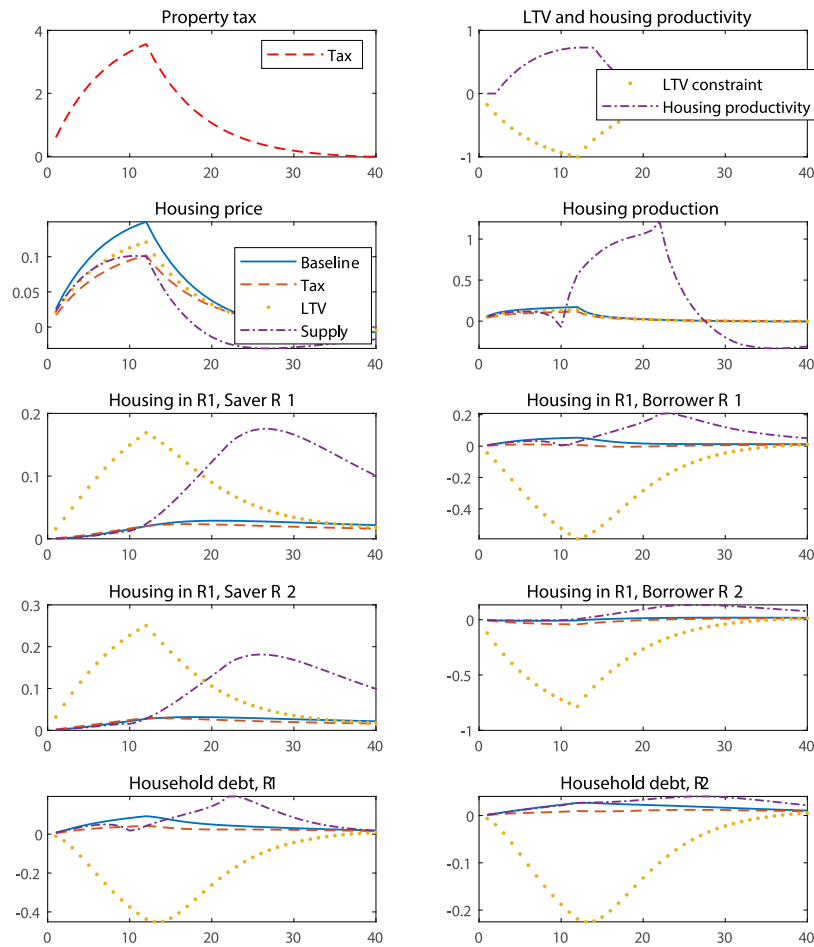


Fig. 5. Stabilization policy, housing exuberance shock in region 1.

Note: Property tax, LTV constraint, and housing productivity are those of region 1. All the variables are expressed as log deviations from the steady state.

long run. Region 1 savers do not change consumption much, but their housing stock declines continuously. Both savers and borrowers in region 1 experience more than a 15 percent decline in housing stocks in 10 years. Lower housing services reduce the utility of all the agents in region 1. Among them, borrowing households in region 1 suffer quicker utility loss because of tightened credit constraints and lower housing services.

In region 2, there is a significant increase in borrowers' consumption, which rises by approximately 0.5 percent. Accordingly, borrowers' utility in region 2 increases. Conversely, there is little change in savers' variables. The credit constraints of region 2 borrowers are much less affected than those of region 1 borrowers because most of the collateral assets held by the former agents are region 2 housing, which is primarily unaffected during the cycle.

5.2. Region-specific stabilization policy

Considering the historical use of those stabilization policy instruments, we examine the effect of the property tax, countercyclical LTV regulation, and housing supply policies on regionally heterogeneous housing cycles. To examine the effects of such policies in our model framework, the strengths of the policy instruments, κ_τ , κ_{LTV} , κ_{hs} , are calibrated such that the adoption of the policy can lower housing prices at the peak by five percentage points. In the case of the LTV policy, we impose an implementable lower bound, as m_t cannot have a negative value, which implies that the cumulative response of $\log(m_t)$ cannot reach below -1.

Figs. 5 and 6 compare the effects of those policy instruments. Fig. 5 shows that it takes quantitatively significant policy responses against

housing demand shocks to reduce housing prices, even by 5 percent. The property tax rate should increase by approximately three times its steady-state value to accomplish this policy effect, which implies that the property tax rate rises from 1 percent to 4 percent. This aggressive increase in the tax rate can be a significant burden for lawmakers. In addition, the figure suggests that a 100 percent reduction in the LTV limit, which leads m_t to reach an implementable lower bound of 0, fails to achieve a 5 percent reduction in housing prices. This inefficacy of the LTV policy is considered to arise from the debt structure in which LTV regulations only apply to new loans, while debt is amortized over time. Meanwhile, to achieve the desired level of policy effect by the supply policy, the government should increase productivity in the housing sector by approximately 50 percent. This approach can also pose challenges for the government in two ways. First, it is unclear whether any housing deregulation can achieve this productivity gain for housing producers. Second, the deregulation of housing, or even the public supply, can cause other social problems (e.g., traffic and environmental issues).

The housing supply policy induces a significant increase in housing production, a rise of approximately 100 percent. Conversely, demand policies such as property taxes and LTV policies slightly reduce housing production. Regarding housing stocks in region 1, the supply policy increases the region 1 housing stock of all agents, while property tax slightly reduces that of all agents compared with the baseline model. The LTV policy increases the region 1 housing of all borrowers and reduces that of all savers in very early periods; however, as the LTV constraint becomes stringent, borrowers reduce their housing stocks, as their ability to borrow becomes limited. Instead, all savers increase their housing purchases.

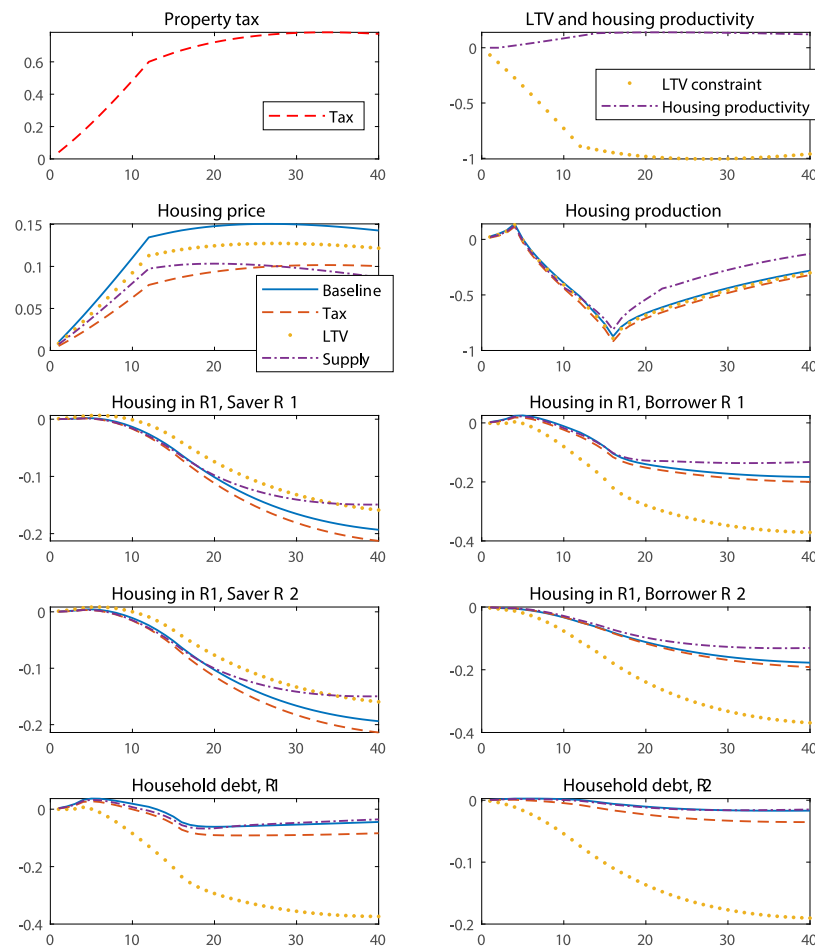


Fig. 6. Stabilization policy, housing supply news shock in region 1.

Note: Property tax, LTV constraint, and housing productivity are those of region 1. All the variables are expressed as log deviations from the steady state.

Household debt in both regions increases by the supply policy as borrowers in both regions can use more region 1 housing stocks as collateral. In contrast, the LTV policy vastly reduces the household debt in both regions as it directly regulates the amount of mortgage lending related to region 1 housing. The tax policy also reduces household debt in both regions, but less than the LTV policy.

Fig. 6 compares the effects of stabilization policies under a regional housing cycle caused by a reduced housing supply. The property tax rate should rise by approximately 0.8 percentage points, and housing supply regulation should be relaxed to increase productivity in the housing sector by approximately 10 percent to decrease housing prices by 5 percent at their peak. Again, the LTV limit reaches an implementable lower bound of zero before achieving the desired policy effect. Housing production further decreases under property tax and LTV regulation but improves under the supply policy.

The policies affect agents' acquisition of region 1 housing in qualitatively similar ways to the demand shock case. Under the supply policy, all types of agents increase region 1 housing stocks. In contrast, the property tax reduces all agents' region 1 housing stocks, as their expected return from housing decreases with higher property tax. The LTV policy increases all borrowers' region 1 housing in the early periods but eventually decreases after the housing cycle passes its peak. All savers' region 1 housing stock shows the opposite pattern. Again, as expected, the LTV policy significantly reduces household debt in both regions.

Table 2 reports a welfare comparison of stabilization policies. The shocks in this experiment arrive unexpectedly for 12 consecutive periods; thus, agents at time 0 cannot account for those shocks. Therefore,

we use a welfare measure that evaluates ex-post utility changes during the cycle, with or without each policy instrument. Welfare is measured by the cumulative change rate of discounted period utility from its steady-state value during 10 years since the beginning of the cycle.

The left panels of Table 2 show that, in the baseline model, region 1 borrowers are better off by the housing demand shock, while region 1 savers are barely affected. In contrast, all agents in region 1 are worse off by the housing supply shock. The critical difference is that the demand shock increases housing stocks in region 1, but the supply shock decreases them to affect housing services for those agents. Especially, borrowers enjoy more welfare gain from the housing demand shock than savers due to the relaxation of credit constraints and more housing services. Given the housing supply shock, low housing service harms region 1 agents' welfare.

Regarding region 2, the housing demand shock decreases the welfare of all region 2 agents while the housing supply shock increases it. Consequentially, both shocks widen the welfare gap between the two regions.

The right panels of the table summarize welfare change induced by each policy compared to the baseline model. There are two remarkable observations from those results. First, the housing supply policy consistently improves the welfare of all agents in region 1. The effect is especially significant against the housing demand shock, where the policy increases savers' welfare by about 10 percentage points and borrowers' welfare by 8 percentage points. The policy is welfare-improving for region 1 as it provides more housing stocks and services; however, the supply policy is welfare-reducing for all agents in region 2. This result is probably because some resources initially available for region 2 are channeled toward housing production in region 1.

Table 2
Welfare analysis of stabilization policies in region 1.

(a) Housing exuberance shock							
	Welfare change (Percent)				Change from the baseline (Percentage points)		
	Baseline	Tax	LTV	Supply	Tax	LTV	Supply
Saver, Region 1	−0.13	−0.44	−0.86	9.42	−0.31	−0.72	9.56
Borrower, Region 1	5.10	3.87	1.24	12.63	−1.23	−3.86	7.53
Saver, Region 2	−0.91	−0.91	−3.94	−1.71	0.00	−3.03	−0.81
Borrower, Region 2	−0.31	0.60	12.32	−3.13	0.91	12.63	−2.82
(b) Housing supply news shock							
	Welfare change (Percent)				Change from the baseline (Percentage points)		
	Baseline	Tax	LTV	Supply	Tax	LTV	Supply
Saver, Region 1	−10.49	−8.20	−6.89	−9.47	2.29	3.60	1.02
Borrower, Region 1	−10.22	−14.91	−24.21	−9.46	−4.69	−13.99	0.76
Saver, Region 2	1.64	2.88	5.59	1.39	1.24	3.95	−0.25
Borrower, Region 2	5.69	5.63	1.88	4.47	−0.06	−3.82	−1.23

Note: “Welfare change” measures the cumulative change rate of discounted period utility from its steady-state value during 40 periods since the beginning of the cycle. “Change from the baseline” reports the difference in welfare change between the baseline model and each policy model.

Second, the LTV policy consistently harms the welfare of region 1 borrowers, as expected. The effect is more significant against the housing supply shock, as the LTV policy exacerbates already tightened credit constraints. The policy affects the region 2 borrowers negatively against the housing supply shock but positively against the housing demand shock.

The welfare effect of the tax policy is unclear. The policy affects region 1 borrowers negatively but less strongly than the LTV policy. Figs. 5 and 6 suggest that the policy suppresses housing production in region 1, suggesting less housing collateral available for region 1 borrowers.

5.3. Monetary policy: leaning against the wind

Because rising housing prices and household debt can pose financial or inflation stability risks, central banks can tighten monetary policy during a housing boom in a leaning against the wind fashion. Here, we discuss the effect of monetary policy when it is used as a stabilization instrument against a regional housing cycle. Whereas the policy instruments in the previous subsection are designed to operate region-specifically, monetary policy is assumed to focus on country-level financial stability. We assume that monetary policy systemically responds to national housing price inflation, where the national housing price index is defined as the weighted average of housing prices in regions 1 and 2. We set the value of r_{π^q} , the response of monetary policy to housing price inflation, to one:

$$R_t = R_{t-1}^R \pi_t^{(1-r_R)r_\pi} (\pi_t^q)^{(1-r_R)r_{\pi^q}} \left(\frac{GDP_t}{G_c GDP_{t-1}} \right)^{(1-r_R)r_Y} \frac{1-r_R}{r} \frac{u_{R,t}}{s_t}, \quad (24)$$

$$\pi_t^q \equiv q_t^i / q_{t-1}^i, \quad q_t^i \equiv \frac{\sum_{j=1}^2 q_{j,t} (h_{j,t} + h'_{j,t} + h_{j,t}^R + h'_{j,t}^R)}{\sum_{j=1}^2 (h_{j,t} + h'_{j,t} + h_{j,t}^R + h'_{j,t}^R)}. \quad (25)$$

Fig. 7 illustrates the effect of leaning against the wind monetary policy against housing demand shocks in region 1. The first finding is that although this degree of response to housing price inflation is not considered weak, monetary policy has a minimal influence on housing price dynamics. This result contrasts with the previous stabilization policy instruments targeted at regional housing markets. Two factors can explain this inefficacy of monetary policy on regional housing prices. First, monetary policy responds to national housing price inflation, not regional. As national housing price inflation is weaker than the regional one, the strength of monetary tightening is also weaker. Second, the effects of monetary policy on households are dispersed

to housing markets in other regions and consumption. Fig. 7 suggests that monetary policy has significant implications for consumption and welfare in both regions, supporting the second channel.

The leaning against the wind monetary policy has asymmetric effects on borrowers and savers. As expected, a higher interest rate during housing appreciations negatively impacts the consumption and housing of both regions' borrowers. In contrast, the income effect from the higher interest rate enables both regions' savers to increase consumption and housing. As a result, this monetary policy improves welfare for savers and reduces welfare for borrowers in all regions.

Fig. 8 shows the effect of leaning against the wind monetary policy against housing supply shocks in region 1. Again, the effect of monetary policy on housing price dynamics is minimal. The monetary policy reduces the consumption and housing of borrowers in both regions and lowers their welfare. As the housing cycle generated by housing supply shocks is persistent, monetary policy tightening and the adverse welfare effects on borrowers are more persistent as well. Monetary policy also slightly reduces savers' consumption during the early periods of the housing cycle; however, savers can increase their housing stock, and their welfare also rises.

To summarize, using monetary policy to stabilize regionally heterogeneous housing booms is ineffective in lowering housing prices and has a spillover effect on other regions that do not experience housing booms. The effect is asymmetric between saving and borrowing households. Monetary tightening by a higher interest rate harms credit-constrained households, especially their consumption, housing, and welfare. In contrast, it benefits saving households in all regions.

6. Robustness analysis

6.1. When only savers invest in inter-regional housing

When households are credit-constrained, like the borrowers in our model, it could be challenging to obtain more credit to invest in inter-regional housing. This subsection examines the robustness of our results in an alternative model where borrowers cannot purchase housing in other regions; thus, only savers can invest in inter-regional housing. We also assume that the ratio of external agent-owned housing to total regional housing remains at 14 percent by increasing the value of J^R to 0.15. Figures A.1 and A.2 in the Appendix show the response of the macroeconomic variables during a regional housing boom caused by housing demand and supply shocks. In these figures, the main findings are consistent with those of the baseline model. The difference

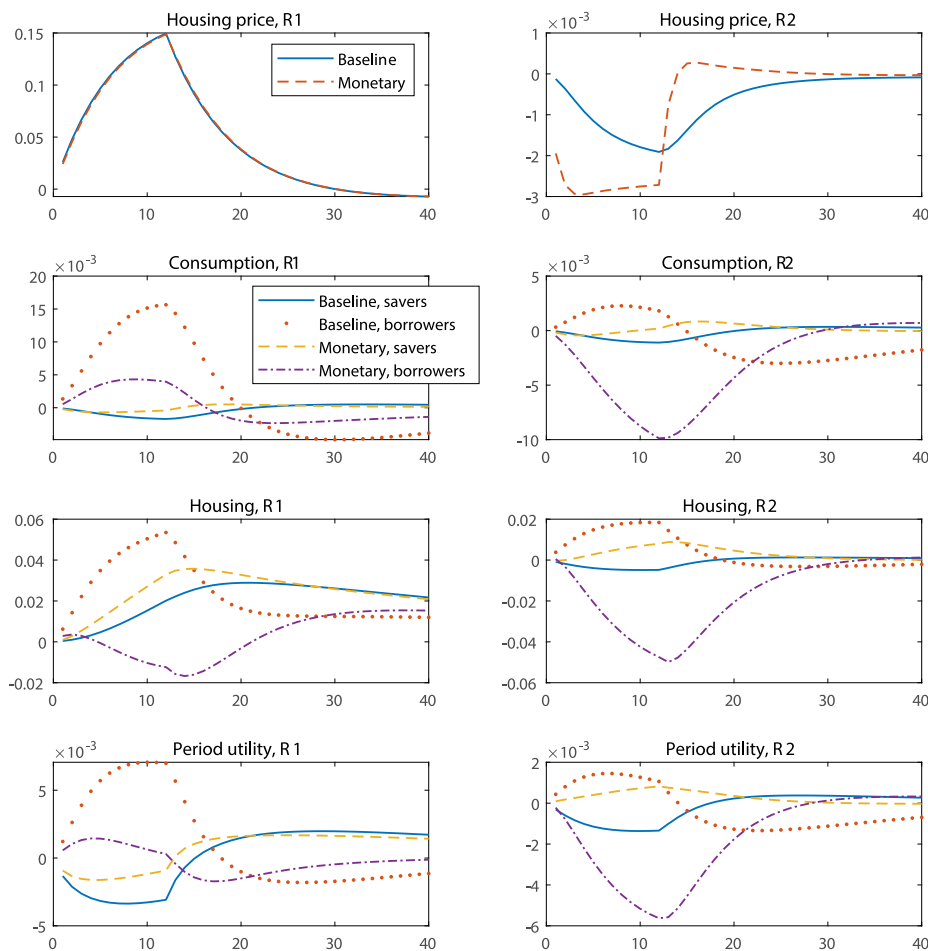


Fig. 7. Leaning against the wind monetary policy, housing exuberance shock in region 1.

Note: All the variables except period utility are expressed as log deviations from the steady state. The period utility is expressed as the percentage change from the steady state.

is that there is less of a spillover effect on the variables in region 2, such as consumption, debt, and welfare. Especially, responses from region 2 borrowers are minimal. This result suggests that inter-regional housing investment by borrowing households, whose motives include speculative ones and ones to enable more access to credit, is vital in the spillover effect of regional housing cycles.

7. Conclusion

This study investigates the macroeconomic implications of regionally heterogeneous housing cycles arising from idiosyncratic shocks in regional housing markets. Their effect on welfare varies by the types of shocks. Increasing demand in the regional housing market can cause regional consumption and housing booms and welfare-improving for borrowers in the region. Rising housing prices in the region because of low housing supply is welfare-reducing for all agents in the region, as they suffer from low housing services. Regional housing shocks also affect agents in other regions, primarily through borrowers' housing investment in the region that experiences the cycles. Therefore, regionally heterogeneous cycles can be a source of economic inequality across regions and levels of financial status.

Regionally heterogeneous housing cycles challenge national policy-makers as welfare implications differ across regions and the type of shocks that cause the cycle. Responding by nationwide policy measures might be inappropriate, as this study's monetary policy is ineffective in stabilizing housing cycles.

This study's analysis enables the comparison of the region-specific housing market stabilization measures that the Korean government

implemented. Among these tools, although the effect varies depending on the source of the housing boom, the housing supply policy provides the best option considering its effect on regional housing prices and welfare. This policy includes deregulations on housing construction or government expenditure to increase the public housing supply; however, caution is needed because the policy can cause other social problems, such as traffic and environmental issues. In contrast, the LTV policy imposes a substantial burden on borrowers in the region experiencing a housing boom.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Simulation program is attached in the supplementary material.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.econmod.2023.106192>.

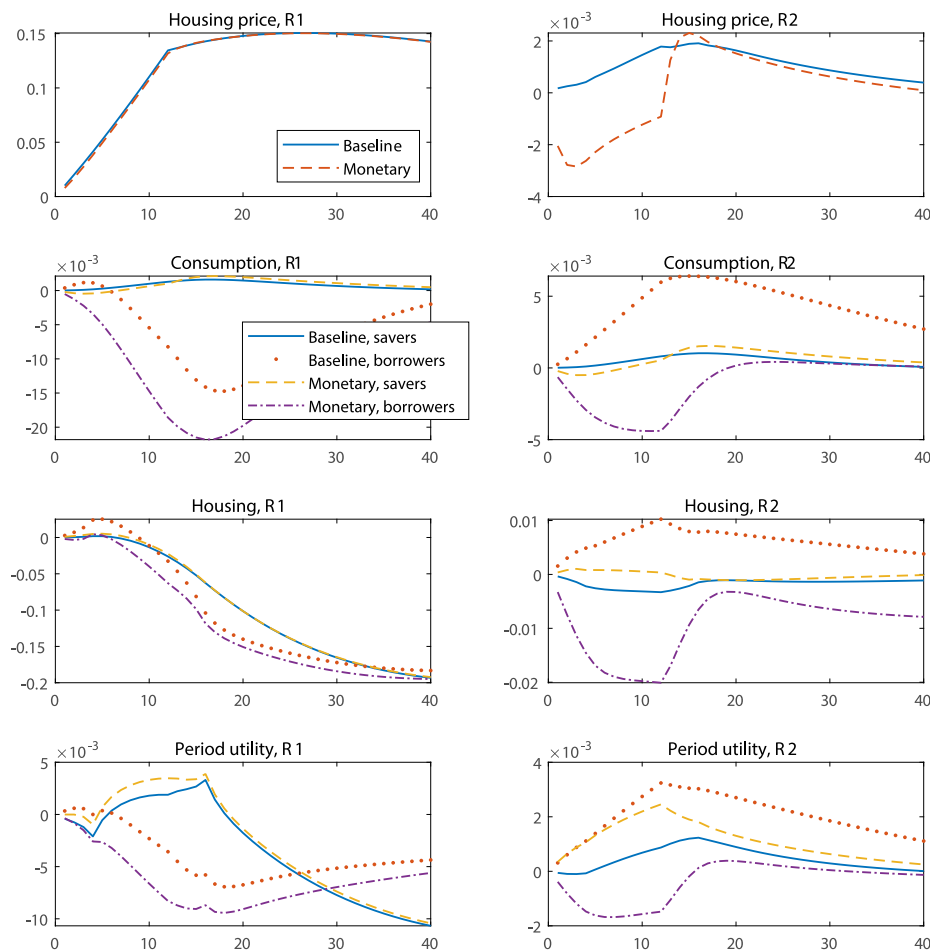


Fig. 8. Leaning against the wind monetary policy, housing supply news shock in region 1.

Note: All the variables except period utility are expressed as log deviations from the steady state. The period utility is expressed as the percentage change from the steady state.

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